
Geov112 Exercise Tectonics- Tiffany

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problem 2

```
% 2a)
[Azi, Velocity]=AziV(56, -28, 7.2, -94, -71);

fprintf('\n')
fprintf('The Nazca plate moving relative to the south America plate \n')
fprintf('\n')
fprintf('The relative motion of the plates is: %s cm/yr \n',
num2str(Velocity))
fprintf('Azimuth: %s degrees', num2str(Azi))
fprintf('\n')
```

```
% 2b)

[Azi1, velocity1]=AziV(62.4, 64.26, 2.1, 135.8, -21.13);

fprintf('\n')
fprintf('The North america side of pingvellir relative to the European
plate \n')
fprintf('\n')
fprintf('The relative motion of the plates is: %s cm/yr \n',
num2str(velocity1))
fprintf('Azimuth: %s degrees', num2str(Azi1+180))
```

Undefined function 'AziV' for input arguments of
type 'double'.

Error in Exercise6_tectonic_Tiffany (line 7)
[Azi, Velocity]=AziV(56, -28, 7.2, -94, -71);

2c)

```
% 25 points along the plate boundary between Africa and South America
% latitude = [0°, -50°]
```

```
points = [
    -0.40, -16.01;
    -1.43, -12.54;
    -3.14, -12.25;
```

```

-6.32, -11.15;
-8.12, -13.15;
-10.38, -13.03;
-13.17, -13.40;
-13.24, -13.40;
-15.51, -13.06;
-18.17, -13.38;
-21.48, -11.46;
-23.21, -13.27;
-27.25, -13.26;
-28.35, -12.33;
-30.19, -13.42;
-31.51, -13.20;
-33.55, -14.36;
-36.12, -17.40;
-38.55, -16.00;
-40.46, -16.47;
-43.38, -16.02;
-45.35, -24.32;
-46.57, -13.25;
-47.55, -10.13;
-48.57, -9.56
];

lat = points(:,1);
lon = points(:,2);

veocity=ones(size(lat));
azis=ones(size(lat));

for i = 1:length(points)

    [beta, v]=AziV(62.5, lat(i), 3.1, -39.4, lon(i));

    veocity(i)= v;
    azis(i) = beta;

end

% subplot;
% Upper plot showing how the velocity varies from 0° to -50°,
% as a sinusoid. The velocity increases and reach its peak around -25°,
% and decreases going futher south from -25°

figure(1)

subplot(2,1,1)
plot(lat, veocity, 'r-*)
title('Velocity for each point')
ylabel('Velocity [cm/yr]')
xlabel('Latitude S ')
grid on

```

```

set(gca, 'XDir', 'reverse')

subplot (2,1,2)
plot(lat, azis, 'm-o')
title('Azimuth for each point')
ylabel('Azimuth °')
xlabel('Latitude S')
grid on
set(gca, 'XDir', 'reverse')

% stephane said to make a plot limited the degrees to 360 for the
% azimuth, showing how little the variation acutall is
figure(2)
subplot (2,1,1)

plot(lat, azis, 'b-o')
title('Azimuth for each point')
ylabel('Azimuth °')
xlabel('Latitude S')
grid on

set(gca, 'XDir', 'reverse')

subplot(2,1,2)

plot(lat, azis, 'm-*')
title('azimuth for each point limited by 360 degrees')
ylabel('Azimuth °')
xlabel('Latitude')
ylim([0,360])
grid on

set(gca, 'XDir', 'reverse')

```

problem 3

```

% 3a)

[m, la, lo]=rotation(13.6,8.7, 55.6, -64.3, -90.1, 96);
fprintf('\n')
fprintf('Antartica relative to the Nazca plate \n')
fprintf('\n')
fprintf('The relative roation of the plates is: %s * 10^-7 deg/yr \n',
num2str(m))
fprintf('The location pole is located at latitude: %s N and longitude:
%s W\n', num2str(la), num2str(abs(lo)))

% 3b)

```

```
% north america-africa:
    % knowing Africa- North America, we can turn the plate position
    % around. going from 78.8 N and 38.3 E => to 78.8 S and 141.7 W
    % angulatt degree: 2.4

% Africa - South America
    % 62.5 N and 39.4 W
    % angular degree: 3.1

% North america - South America
[m1, la1, lo1]=rotation(2.4,3.1, -78.8, 62.5, -141.7, -39.4);

fprintf('\n')
fprintf('North America plate relative to the South America plate \n')
fprintf('\n')
fprintf('The relative roation of the plates is: %s * 10^-7 deg/yr \n',
num2str(m1))
fprintf('Latitude: %s N and Longitude: %s W \n', num2str(la1),
num2str(abs(lo1)))
```

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